MAURICE BARAZA OTIENO

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EDUCATION

Lafayette College **Easton, PA**
Bachelor of Science Electrical and Computer Engineering **Graduation: May 2026**
Relevant Coursework: Mixed Signal-Design of electronic circuits, Signals and Communication Systems, Digital circuits and computer Architecture, Embedded Systems, Industrial Electronics and Control systems, Electromagnetics, Power Electronics, Data Structures and Algorithms, Artificial Intelligence for computer vision, Speech and Image processing

Relevant Skills: Design with FPGA(System Verilog HDL design and Verification), MCUs, mixed-signal design, UVM, circuit simulation (LTSpice, Ngspice, PSPICE), PCB design, KiCad, Altium Designer, Cadence ORCADX, Bench testing, Soldering & rework, Embedded design, Python scripting, LINUX shell, FreeRTOS, BareMetal C/C++, STM32CUBE, UART, I^2C, SPI, ADC, DACs, RASPBERRY PI PICO, Motor Control, DC-DC convertors, MATLAB, Visual Studio

Awards and Honors: Murray G. Clay '30 Award Fund in Engineering Science, Dean's List: 2 semesters
Campus Engagement: IEEE, Lafayette African and Caribbean Students Association, Soccer club, and National Society of Black Engineers.

RELEVANT EXPERIENCE

Kenya Power and Lighting Company: *Junior Intern, Migori, Kenya* **06/ 2025 – 08/ 2025**
 Participated in preventive maintenance and inspection of distribution transformers to ensure system reliability and safety compliance.
 Actively involved in the replacement of faulty electricity distribution transformer components under supervision.

Selected Project Experience ***[All projects and personal information will be found at* https://mauriceotieno.com]**

Smart Technology Rollator for Independence Detection and Empowerment (STRIDE) -Autonomous Stair Climbing system **08/2025 - Present**
Designed the power management system incorporating a 24V battery pack, current monitoring, and protection circuitry for safe motor operation.
Designed the low-level embedded control architecture integrating STM32 microcontrollers, BLDC motor drivers, and sensor feedback for closed-loop mobility control using the CMSIS V2 FreeRTOS in STM32CUBEIDE.
 Collaborated with seven teammates to perform hardware integration, PCB design, and system-level debugging across analog, digital, and power subsystems.

Pulse Width Modulation Signal generating Voltage Controlled oscillator **01/2025– 3/2025**
 Designed a PWM-generating VCO based on a charge/discharge cycle of a capacitor using analog IC op amps, matched CMOS transistors as current mirrors, comparators, and RS NAND latches.
 Simulated the design's performance in LTSPICE and implemented the design on a printed circuit board via a schematic-driven layout in KICAD.
 Performed hardware bring-up and debug using an oscilloscope and function generator, correlating measured performance with simulation
https://github.com/otienomaurice/VOLTAGE_CONTROLLED_OSCILLATOR.git

Digital Design: Pulse Sensor and Reaction Timer **10/2022-12/2022**
 Collaborated with a partner to Design and implement a System Verilog hardware module for pulse detection and reaction-time measurement on a Nexys A7100T FPGA.
 Verified functional correctness and timing behavior through waveform simulation and testbench development in Vivado
 Integrated and debugged the design on hardware via PMOD interfaces, resolving timing and synchronization issues.
.https://github.com/otienomaurice/Pulse_Sensor_and_reaction_timer-.git

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Equity Group Foundation: *Intern, College Counselling* **05/2022-08/2022**
 Mentored recent high school graduates on academic planning, university applications, and career development.
 Provided structured guidance sessions focused on goal setting, scholarship preparation, and transition to tertiary education.

ADDITIONAL EXPERIENCE

ECE Research Fellow, Lafayette College **06/ 2025-06/2026**
Student IT Assistant, Tech Lounge, Lafayette College **01/ 2024-06/2026**
Student Library Assistant, Lafayette College **08/2023-06/2026**
Languages: English (Fluent), Swahili (Fluent), Luo (Fluent), Sheng (Conversational)